

CIRROSE NA UTI

PARTE 3



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CONTINUA...

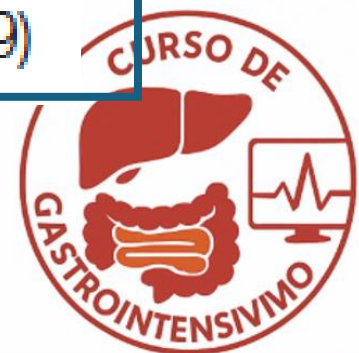
HDA VARICOSA



SUSPEITA CLÍNICA!

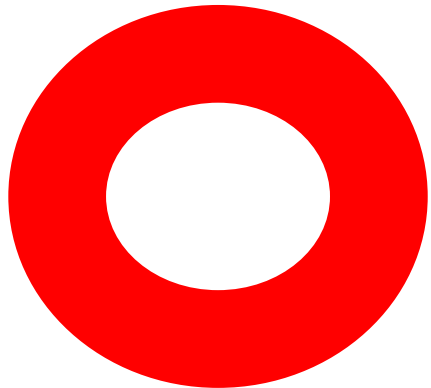
ALTA X BAIXA

		Specificity, % (95% CI)
Conteúdo Gástrico (sangue/"borra de café").	HDA	95 (90-98)
Melena (visto pelo médico)	HDA	98 (91-99.6)
Coágulos (visto pelo médico)	HDB	99.2 (96.0-99.9)
Relação Ur/Cr >63 (BUN 30)	HDA	93 (87 to 99)



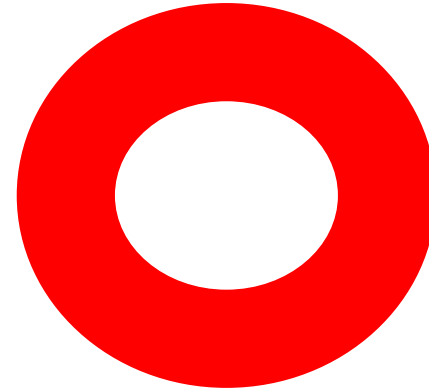
CONTROLE HB ?

Fase Inicial
Hb mantido?



Perda de Eritrócitos + Plasma

Fase tardia
Hb Cai



Plasma



Fluidos



DIAGNÓSTICO

Endoscopia – Padrão Ouro

- Até 24 H – reduz mortalidade
- Visualização Direta (Etiologia)
- Terapêutica



Quando IOT p EDA



Sedação



Broncoaspiração
Hipotensão
RNC
PCR

Laine, L. (2016). "Upper Gastrointestinal Bleeding Due to a Peptic Ulcer." *N Engl J Med* **375**(12): 1198.



EDA É EMERGÊNCIA OU PODE ESPERAR ESTABILIZAR ?

Timing of Endoscopy for Acute Upper Gastrointestinal Bleeding

James Y.W. Lau, M.D., Yuanyuan Yu, Ph.D., Raymond S.Y. Tang, M.D., Heyson C.H. Chan, M.B., Ch.B., Hon-Chi Yip, M.B., Ch.B., Shannon M. Chan, M.B., Ch.B., Sally W.Y. Luk, M.B., Ch.B., Sunny H. Wong, Ph.D., Louis H.S. Lau, M.B., Ch.B., Rashid N. Lui, M.B., Ch.B., Ting T. Chan, M.B., Ch.B., Joyce W.Y. Mak, M.B., Ch.B., Francis K.L. Chan, M.D., and Joseph J.Y. Sung, M.D.

N ENGL J MED 382;14 NEJM.ORG **APRIL 2, 2020**

Mortalidade 30 dias

P 0.34



8,9%

EDA < 6h



6,6%

EDA 6-24h



SUORTE HEMODINÂMICO X HDA

Gut, 1974, 15, 99-101

The portal pressure-blood volume relationship in cirrhosis¹

DAVID S. ZIMMON AND RICHARD E. KESSLER

From the Medical and Surgical Services, Veterans Administration Hospital, and New York University School of Medicine, New York, New York

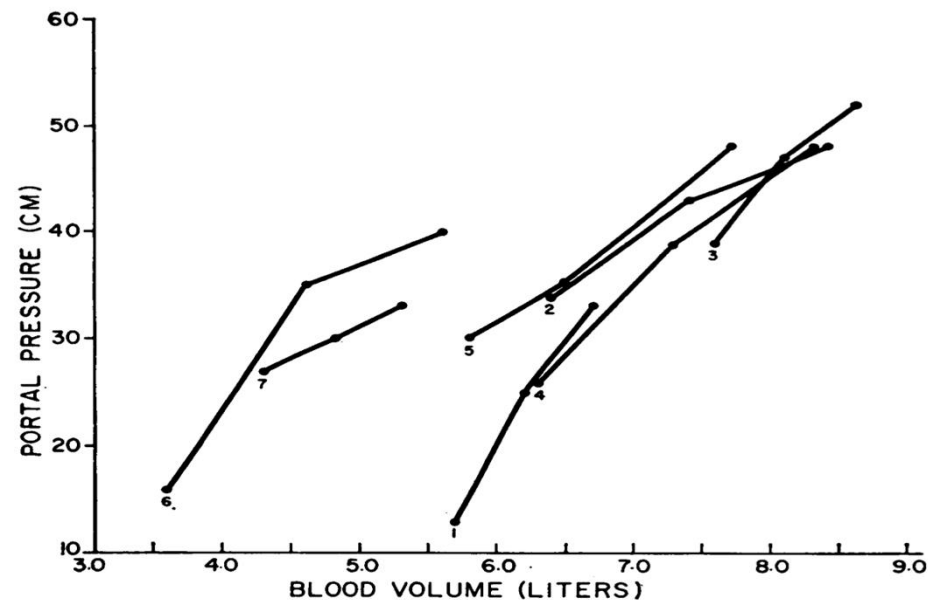


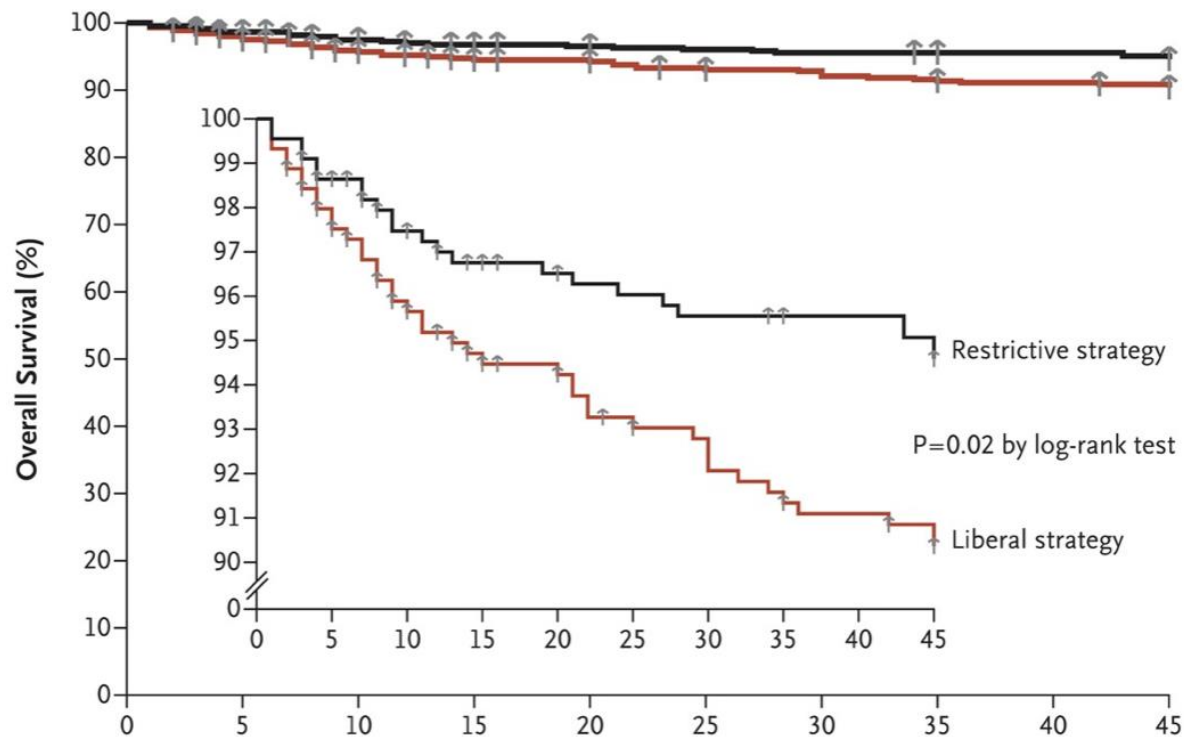
Fig. 1 *Portal pressure-blood volume curves from seven cirrhotic patients with portal hypertension derived from three measurements of portal pressure at blood volumes differing by more than 500 ml.*

- PAM > Vasopressores
- Expansão x Pressão Portal



SUORTE TRANSFUSIONAL ?

A Survival, According to Transfusion Strategy

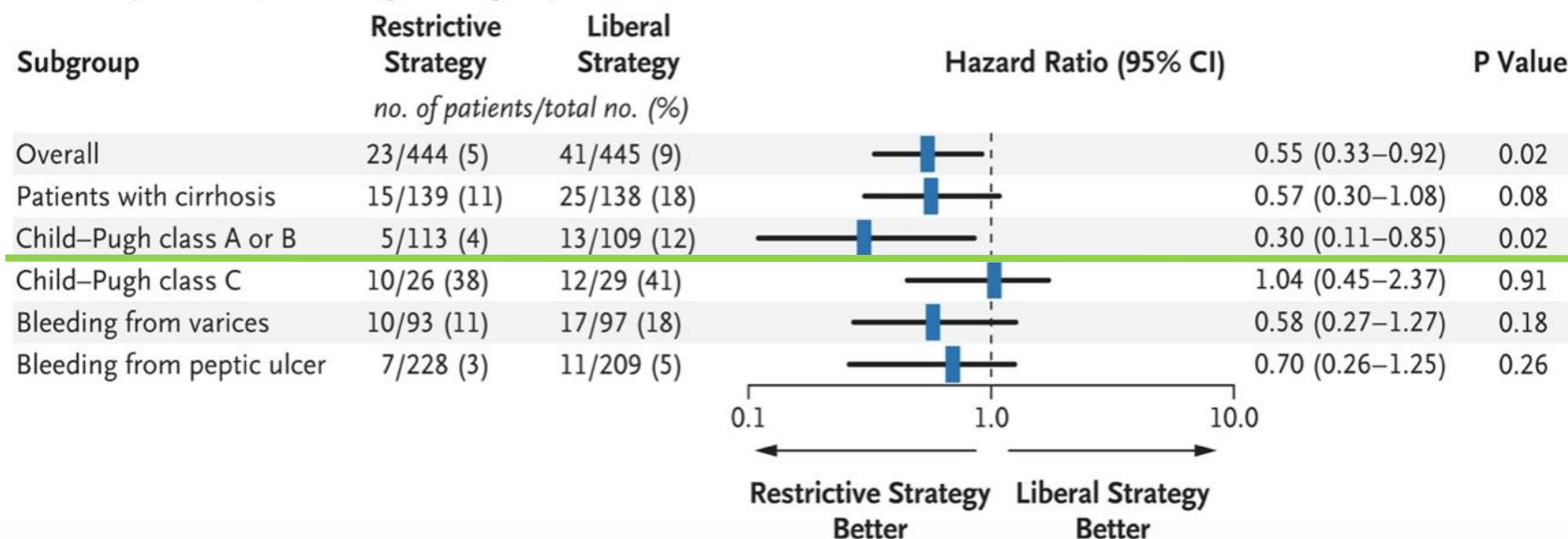


Alvos Hb 7-9 x > 9

- Menor Mortalidade
- Menor Sangramento
- Menor Pressão Portal



B Death by 6 Weeks, According to Subgroup



0.45 to 2.37). Within the first 5 days, the portal-pressure gradient increased significantly in patients assigned to the liberal strategy ($P=0.03$) but not in those assigned to the restrictive strategy.



PROFILAXIA DE TROMBOSE

REVIEW

Perioperative thromboprophylaxis in liver transplant patients

NEW CONCEPT OF REBALANCED HEMOSTASIS AND THE UTILITY OF STANDARD LABORATORY TESTS

Table 1 Balance of antihemostatic and prohemostatic drivers in the cirrhotic patient

	Anti-hemostatic drivers	Pro-hemostatic drivers
Primary hemostasis	Abnormal platelet function Thrombocytopenia	Elevated vWF Reduced ADAMTS 13
Coagulation	Decreased production of thrombopoietin Reduced synthesis of factors II, V, VII, IX, X, and XI Vitamin K deficiency	Platelet hyperreactivity Elevated factor VIII Reduced anticoagulant protein C, protein S, antithrombin III
Fibrinolysis	Hypo-dysfibrinogenemia Low α 2-antiplasmin, factor XIII, and TAFI Elevated tPA	Procoagulant changes in fibrin structure Low plasminogen High PAI



ROTEM in the setting of liver transplant surgery reduces frozen plasma transfusion

Muntadhar Al Moosawi^a, Jacqueline Trudeau^{a,b}, Tyler Smith^{a,c}, Alexandre Lefebvre^d, Andrew W. Shih^{a,c,*}

^a Department of Pathology and Laboratory Medicine, University of British Columbia, Vancouver, British Columbia, Canada

^b Department of Anesthesiology, Pharmacology, and Therapeutics, University of British Columbia, Vancouver, British Columbia, Canada

^c Centre for Blood Research, University of British Columbia, Vancouver, British Columbia, Canada

^d Department of Cardiothoracic Anesthesiology, Copenhagen University Hospital, Copenhagen, Denmark

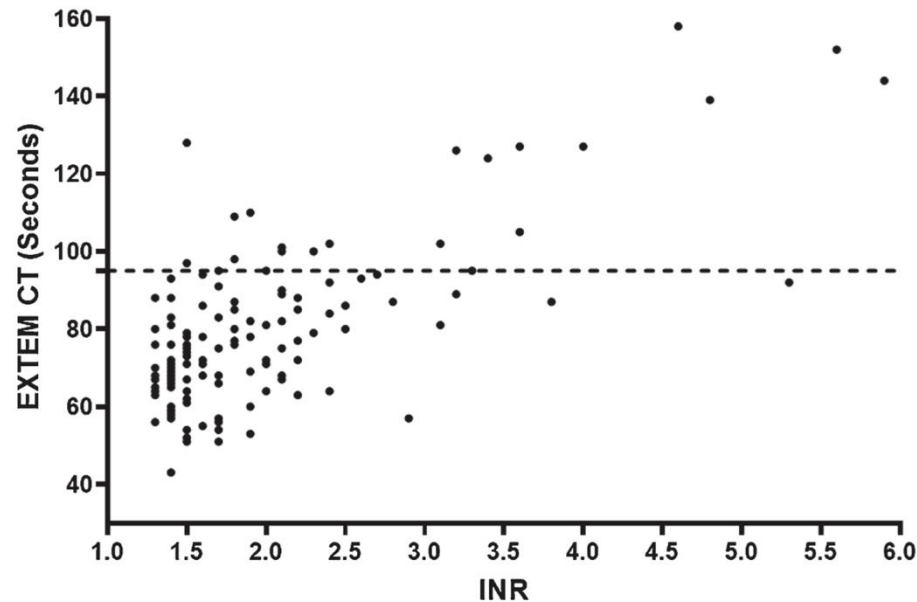


Fig. 1. Correlation between INR and EXTEM CT.

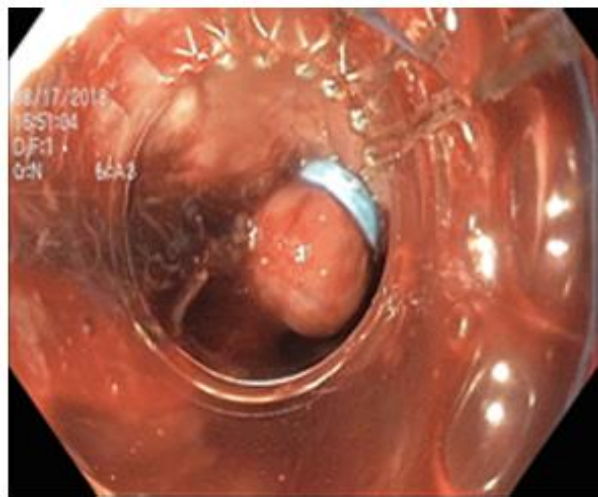
The horizontal line represents the upper limit of normal for the clotting time using the EXTEM reagent based on reference ranges used at our center.

The correlation between INR
and EXTEM CT improves only when INR is greater than 3.0 ($\rho = 0.703$).



TRATAMIENTO ENDOSCÓPICO

- › VE - Ligadura
- › GOV - Cianoacrilato
- › EDA controle ?



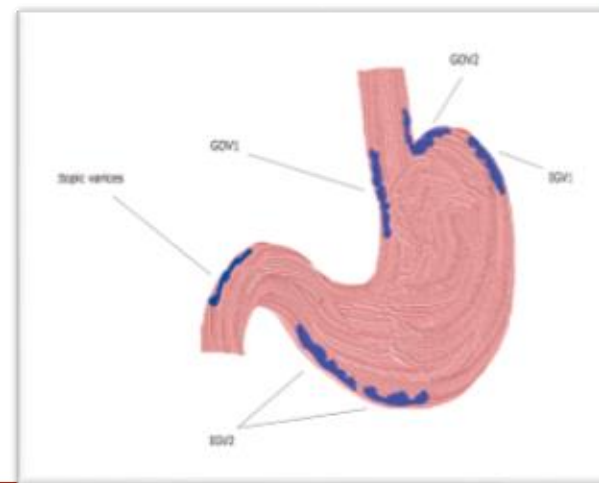
Ligadura de VE sangrante

<https://www.wjnet.com/1948-5182/figures/v11/i3/250.htm>

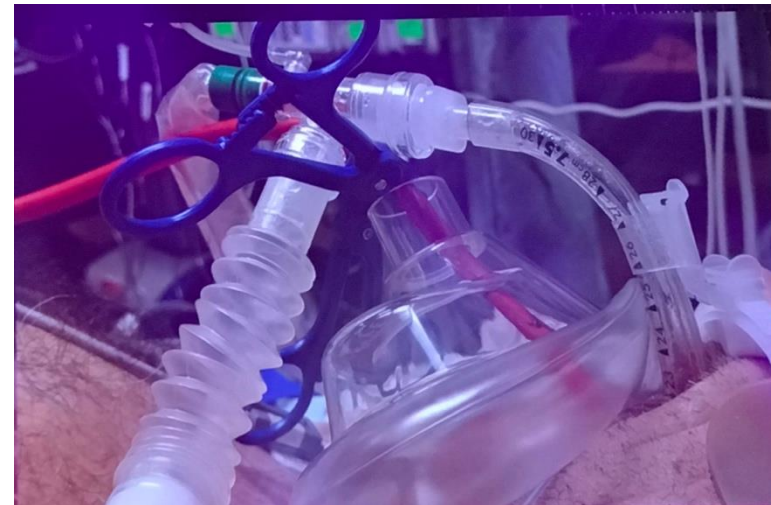
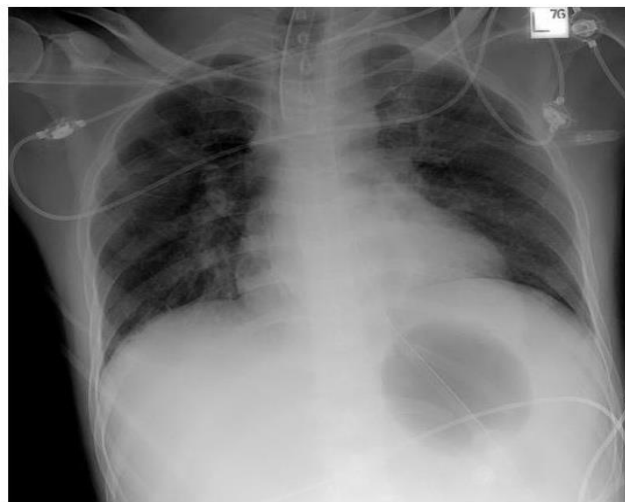
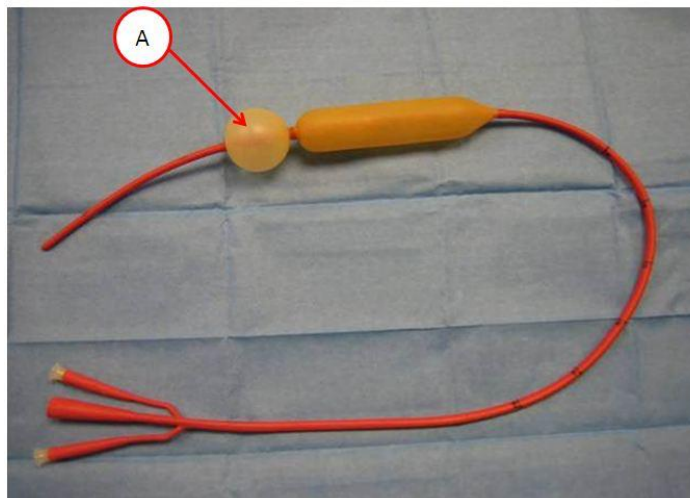
Table 6. Classification, prevalence and risk of bleeding of gastric varices.

Type	Definition	Relative frequency	Overall bleeding risk without treatment
GOV 1	OV extending below cardia into lesser curvature	70%	28%
GOV 2	OV extending below cardia into fundus	21%	55%
IGV 1	Isolated varices in the fundus	7%	78%
IGV 2	Isolated varices else in the stomach	2%	9%

GOV, gastro-oesophageal varices; IGV, isolated gastric varices; OV, oesophageal varices.



BALÃO DE TAMPONAMENTO GASTROESOFÁGICO



Proteger Via
aérea



Checar/lubrifi
car Sonda



Posicionar
45°



Medir =
SNG/SOG



Passagem
Oral/Nasal



Fixar com
Tração



Inflar
Esofágico SN



Tracionar – Rx



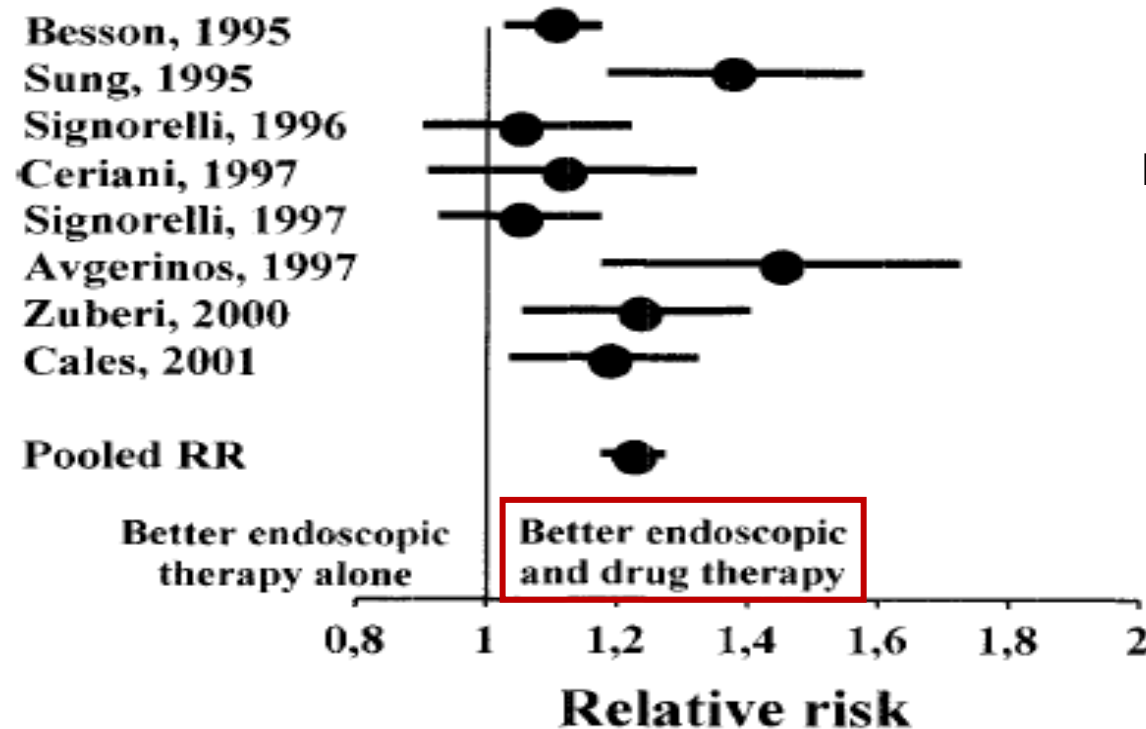
Inflar gástrico
completo



Confirmar
com Rx

VASOCONSTRITOR ESPLÂNCNICO

Terlipressina / Somatostatina / Octreotide
2-5 dias



Melhor controle de sangramento !
Mortalidade ????

Banares, R., et al. (2002). *Hepatology* 35(3): 609-615.



VASOCONSTRICTOR ESPLÂNCNICO >> TANTO FAZ ?

Lack of Difference Among Terlipressin, Somatostatin, and Octreotide in the Control of Acute Gastroesophageal Variceal Hemorrhage

Table 2. Treatment Responses in All Enrolled Patients With Variceal Bleeding

Treatment Response	All Patients (n = 780)	Terlipressin Group (n = 261)	Somatostatin Group (n = 259)	Octreotide Group (n = 260)	P Value
Control of index bleeding without rescue therapy, n (%)	690 (88.5)	234 (89.7)	227 (87.6)	229 (88.1)	0.752
Time interval from IO to bleeding control, hours	9.8 ± 10.3	9.6 ± 10.7	10.1 ± 10.1	9.7 ± 10.2	0.839
Time interval from commencement of vasoactive drug to bleeding control, hours	8.0 ± 10.3	7.8 ± 10.6	8.2 ± 10.1	8.1 ± 10.3	0.899
Patients with rebleeding, n (%)*	29 (4.2)	8 (3.4)	11 (4.8)	10 (4.4)	0.739
Time interval from IO to rebleeding, hours	66.8 ± 23.6	70.0 ± 27.5	69.0 ± 19.6	61.9 ± 26.0	0.730
Time interval from bleeding control to rebleeding, hours [†]	62.1 ± 24.4	64.7 ± 26.4	64.2 ± 22.4	57.9 ± 26.8	0.801
Mortality, n (%)	67 (8.6)	21 (8.0)	23 (8.9)	23 (8.8)	0.929
Cause of mortality					0.920
Uncontrolled index bleeding, n (%)***	60 (89.6)	19 (90.5)	20 (87.0)	21 (91.3)	
Uncontrolled rebleeding, n (%) [‡]	2 (3.0)	1 (4.8)	1 (4.3)	0 (0)	
Liver failure, n (%) [‡]	3 (4.5)	1 (4.8)	1 (4.3)	1 (4.3)	
Infection, n (%) [‡]	2 (3.0)	0 (0)	1 (4.3)	1 (4.3)	
5-day treatment success, n (%)	659 (84.5)	225 (86.2)	216 (83.4)	218 (83.8)	0.636

- Terlipressina 2mg 4/4 >> 1mg 4/4h
- Somatostatina 250 µg bolus >> 250 µg/h
- Octreotide 50µg bolus >> 50 µg/h



A randomized controlled trial comparing octreotide and vasopressin in the control of acute esophageal variceal bleeding

Journal of Hepatology, 1992; 16: 320–325

Shinn-Jang Hwang, Han-Chieh Lin, Chia-Fu Chang, Fa-Yauh Lee, Chi-Wen Lu, Hsiao-Chung Hsia, Sun-Sang Wang, Shou-Dong Lee, Yang-Te Tsai and Kwang-Juei Lo

Division of Gastroenterology, Department of Medicine, Veterans General Hospital and National Yang-Ming Medical College, Taipei, Taiwan, Republic of China

(Received 23 August 1991)

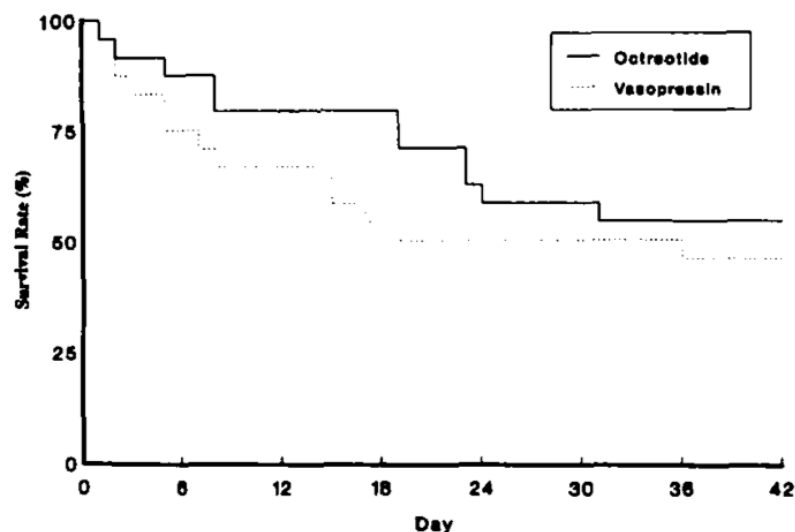


Fig. 1. The survival rate after acute variceal bleeding showed no significant difference in the patients treated with octreotide (solid line) and vasopressin (broken line).

TABLE 3

Bleeding control rate in cirrhotic patients with or without HCC

	<i>n</i>	Initial control		Complete control	
		<i>n</i>	%	<i>n</i>	%
Octreotide					
HCC(+)	7	5	71	3	43
HCC(-)	17	16	94*	12	71
Total	24	21	88**	15	63***
Vasopressin					
HCC(+)	7	2	29	2	29
HCC(-)	17	11	65*	9	53
Total	24	13	54**	11	46***

* $p = 0.08$; ** $p = 0.01$; *** $p > 0.05$ HCC = hepatocellular carcinoma.



A Prospective Randomized Controlled Clinical Trial
Comparing the Effects of Somatostatin and
Vasopressin for Control of Acute Variceal Bleeding in
the Patients with Liver Cirrhosis

Heon Young Lee, M.D., Heon Ju Lee, M.D.*, Seung Min Lee, M.D.,
Jun Hwan Kim, M.D.*, Soon Wook Kweon, M.D.*,
Byung Seok Lee, M.D. and Nam Jae Kim, M.D.

Table 3. Results of Vasopressin and Somatostatin treatments for
the control of acute variceal bleeding

	Successful	Not Successful	p-Value
during initial 6 hours			
Vasopressin (n=23)	20	3	0.37
Somatostatin (n=20)	19	1	
from 6 to 120 hours			
Vasopressin (n=20)	18	2	0.35
Somatostatin (n=19)	15	4	

Dose 0.4 Ui/min

Table 5. Complications of treatments

	Vasopressin (n=23)	Somatostatin (n=20)
Abdominal pain or Chest pain	5	0 ^Ω
Bradycardia	3	2
Elevated blood pressure	2	4

Ω, p<0.05





Low-dose continuous terlipressin infusion is effective and safer than intravenous bolus injections in reducing portal pressure and control of acute variceal bleeding

Vinod Arora¹ · Shakti Prasad Choudhary¹ · Rakhi Maiwall¹ · Rajan Vijayaraghavan¹ · Ankur Jindal¹ · Guresh Kumar² · Shiv Kumar Sarin¹



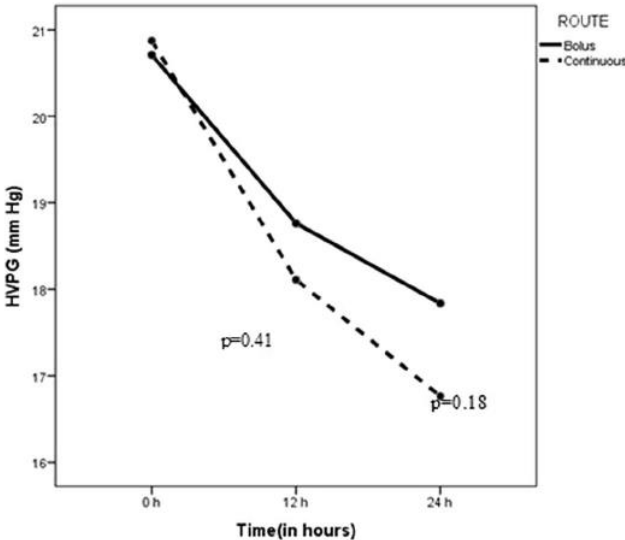
Bolus: 2mg 4/4h



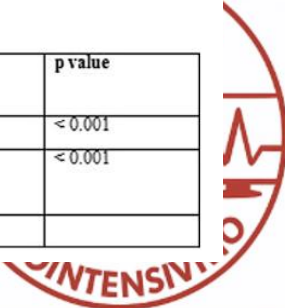
Infusão Contínua: 4mg/24h

Eficácia =

Fig. 2 Significant reduction was noted in HVPg in both the arms compared to the baseline pressure ($p < 0.001$) however, between the BOL and CONI, no significant difference was noted in the HVPg levels at 12 ($p = 0.41$) and 24 h ($p = 0.18$)



Terlipressin administration	HVPg0	HVPg12	HVPg24	p value
Bolus Terlipressin (n=55)	20.71 ± 3.73	18.76 ± 4.03	17.84 ± 4.20	< 0.001
Continuous Terlipressin Infusion (n=55)	20.87 ± 3.47	18.11 ± 4.22	16.76 ± 4.15	< 0.001
p value	0.81	0.41	0.18	





Low-dose continuous terlipressin infusion is effective and safer than intravenous bolus injections in reducing portal pressure and control of acute variceal bleeding

Vinod Arora¹ · Shakti Prasad Choudhary¹ · Rakhi Maiwall¹ · Rajan Vijayaraghavan¹ · Ankur Jindal¹ · Guresh Kumar² · Shiv Kumar Sarin¹ 



Bolus: 2mg 4/4h



Infusão Contínua: 4mg/24h

< Efeito colateral!

Hepatology International

Table 3 a Adverse events related to terlipressin administration in the treatment groups. b. Adverse events leading to discontinuation of drug

Variable	Bolus terlipressin administration	Continuous terlipressin Infusion	p value
Diarrhea (more than 4 times/day)	6/55 (10.9%)	5/55 (9.09%)	1.00
Abdominal pain	4/55 (7.2%)	2/55 (3.6%)	0.67
Atrial fibrillation	3/55 (5.4%)	2/55 (3.6%)	1.00
Cyanosis	5/55 (9.09%)	2/55 (3.6%)	0.43
Chest pain	3/55 (5.4%)	2/55 (3.6%)	1.00
Ventricular premature complex (VPCs)#	4/55 (7.3%)	2/55 (3.6%)	0.67
Hypertension	6/55 (10.9%)	5/55 (9.09%)	1.00
Total	31/55 (56.4%)	20/55 (36.3%)	0.03
Variable	Bolus terlipressin administration	Continuous terlipressin infusion	p value
Diarrhea (more than 4 times/day)	2/55 (3.6%)	1/55 (1.8%)	1.00
Abdominal pain	1/55 (1.8%)	1/55 (1.8%)	1.00
Atrial fibrillation #	2/55 (3.6%)	1/55 (1.8%)	1.00
Cyanosis	3/55 (5.4%)	1/55 (1.8%)	0.61
Chest pain	2/55 (3.6%)	1/55 (1.8%)	1.00
Ventricular premature complex (VPCs)#	1/55 (1.8%)	–	
Hypertension	3/55 (5.4%)	–	
Drug discontinuation	14/55 (25.4%)	5/55 (9%)	0.02



Effects of a high-dose 24-h infusion of tranexamic acid on death and thromboembolic events in patients with acute gastrointestinal bleeding (HALT-IT): an international randomised, double-blind, placebo-controlled trial

TRANSAMIN ?

The HALT-IT Trial Collaborators*

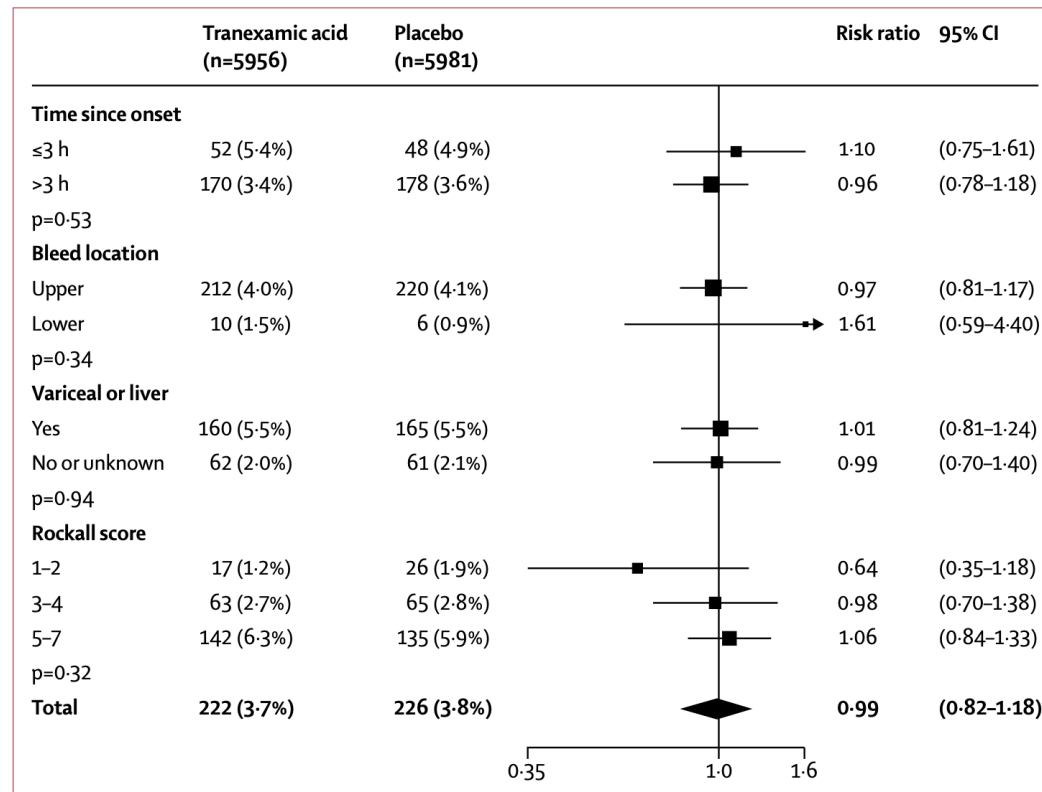


Figure 3: Effect of tranexamic acid on death due to bleeding within 5 days
Analysis stratified by time since bleeding onset, suspected bleed location, suspected variceal bleeding or comorbid liver disease, and Rockall score.

Lancet 2020; 395: 1927-36



Effects of a high-dose 24-h infusion of tranexamic acid on death and thromboembolic events in patients with acute gastrointestinal bleeding (HALT-IT): an international randomised, double-blind, placebo-controlled trial

*The HALT-IT Trial Collaborators**

	Tranexamic acid	Placebo	Outcomes
Complications			
Any thromboembolic event	86/5952 (1.4%)	72/5977 (1.2%)	1.20 (0.88 to 1.64)
Venous events (deep vein thrombosis, pulmonary embolism)	48/5952 (0.8%)	26/5977 (0.4%)	1.85 (1.15 to 2.98)

Lancet 2020; 395: 1927–36



Tranexamic acid in upper gastrointestinal bleed in patients with cirrhosis - a randomized controlled trial

Kumar, Manoj¹; Venishetty, Shantan¹; Jindal, Ankur¹; Bihari, Chhagan²; Maiwall, Rakhi¹; Vijayaraghavan, Rajan¹; Saggere Muralikrishna, Shasthry¹; Arora, Vinod¹; Kumar, Guresh³; Sarin, Shiv K.¹

Author Information

Hepatology ():10.1097/HEP.0000000000000817, March 5, 2024. | DOI: 10.1097/HEP.0000000000000817

Table 2. Clinical course and mortality of patients with acute upper gastrointestinal bleed receiving tranexamic acid and placebo.

Parameters	Tranexamic acid group (n=300)	Placebo group (n=300)	Risk estimate (95% CI)*	P value
Underwent pre-emptive TIPS or TVOGV after first endotherapy	5 (1.6%) [4- TIPS; 1- TVOGV]	4 (1.3%) [3-TIPS; 1- TVOGV]	1.250 (0.339 to 4.610)	0.736
Failure to control bleeding by day 5	19(6.3%)	40(13.3%)	0.457 (0.282 to 0.801)	0.006



Tranexamic acid in upper gastrointestinal bleed in patients with cirrhosis - a randomized controlled trial

Kumar, Manoj¹; Venishetty, Shantan¹; Jindal, Ankur¹; Bihari, Chhagan²; Maiwall, Rakhi¹; Vijayaraghavan, Rajan¹; Saggere Muralikrishna, Shasthry¹; Arora, Vinod¹; Kumar, Guresh³; Sarin, Shiv K.¹

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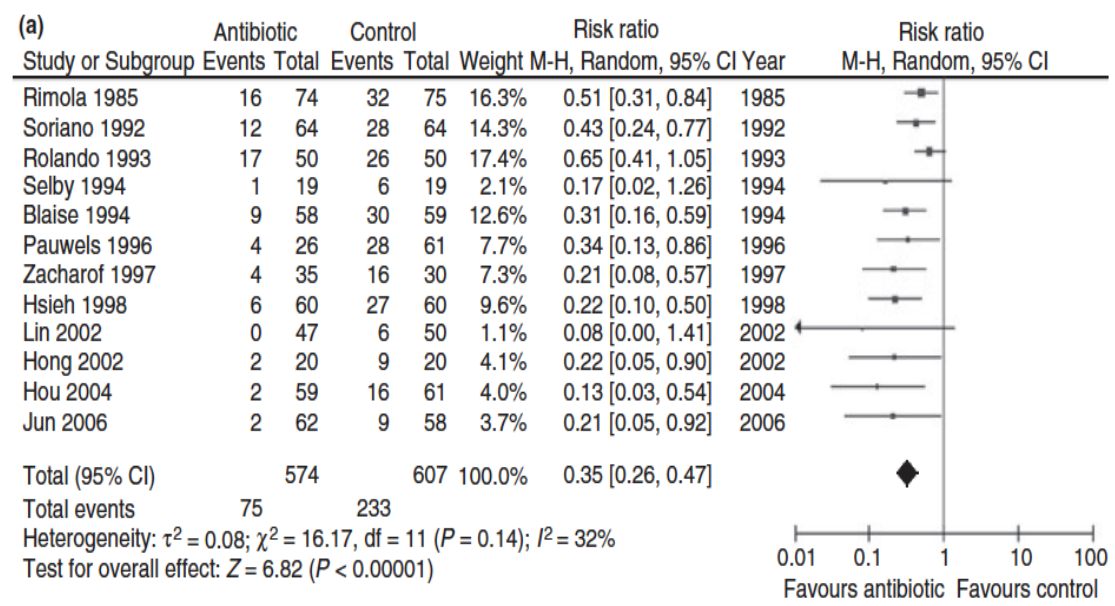
Table 3. Side effects and new complications in patients with acute upper gastrointestinal bleed receiving tranexamic acid and placebo.

Parameters	Tranexamic acid group (n=300)	Placebo group (n=300)	Risk ratio (95% CI)	P value
Any thromboembolic event	1 (0.3%)	0	1.003(0.997 to 1.010)	1.00
Arterial events	1 (0.3%)	0	1.003(0.997 to 1.010)	1.00
Myocardial infarction	0	0	-	-
Stroke	0	0	-	-
Transient ischemic attack	1 (0.3%)	0	-	1.0
			1.003(0.997 to 1.010)	
Venous event	0	0	-	-
Deep vein thrombosis	0	0	-	-
Pulmonary embolism	0	0	-	-
Other cardiac events	0	1 (0.3%) [Atrial fibrillation]	0.997 (0.990 to 1.003)	1.0

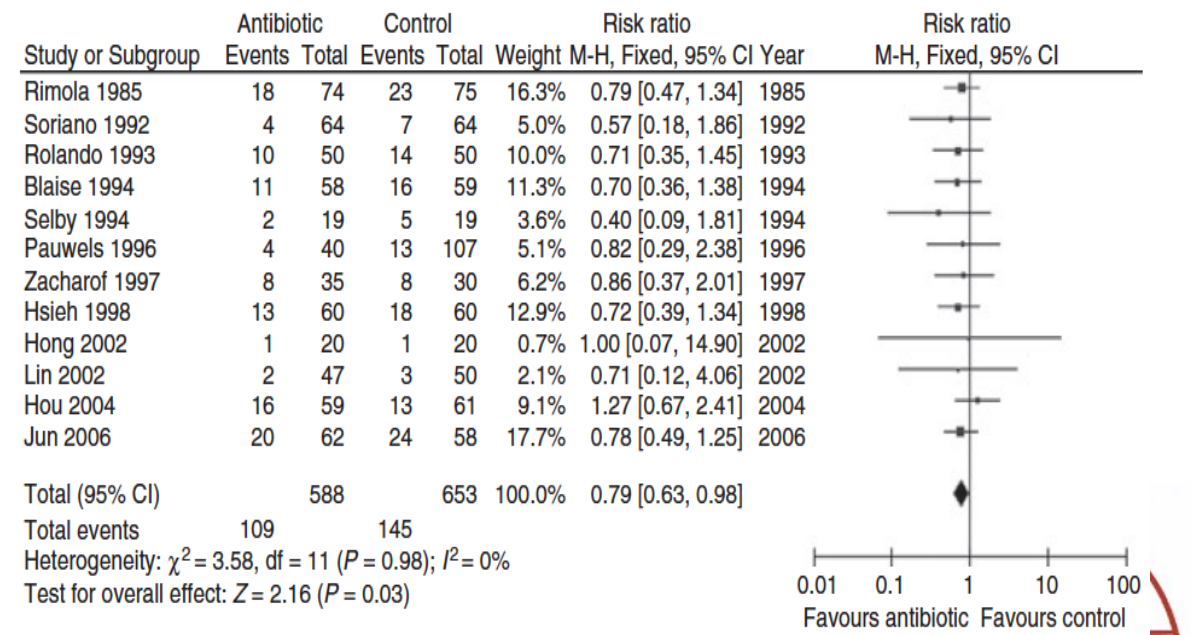


ANTIBIOTIC PROFILAXIA X HDA

Redução de Infecções ?



Redução de Mortalidade ?



Prophylactic Antibiotics for Upper Gastrointestinal Bleeding in Patients With Cirrhosis

A Systematic Review and Bayesian Meta-Analysis

Connor Prosty, MD; Dean Noutsios, MD; Laurie-Rose Dubé, MD; Rachel Baden, MD; Kusha Davar, MD; Sarah Freling, MD; Taft Bhuket, MD; Hal F. Yee Jr, MD, PhD; Brad Spellberg, MD; Emily G. McDonald, MD, MSc; Todd C. Lee, MD, MPH

JAMA Intern Med. doi:10.1001/jamainternmed.2025.3832

Published online August 11, 2025.

defined, and no studies reported adverse events. Two RCTs compared longer (5-7 days) to shorter (2-3 days) durations and 12 RCTs compared any prophylaxis (1-10 days) to none. Shorter durations (including none) had a 97.3% probability of noninferiority for all-cause mortality (RD, 0.9%; 95% credible interval [95% CrI], -2.6 to 4.9). Shorter durations had a 73.8% probability of noninferiority for early rebleeding (RD, 2.9%; 95% CrI, -4.2 to 10.0) but were associated with more study-defined bacterial infections (RD, 15.2%; 95% CrI,



Antibiotic Prophylaxis in Variceal Hemorrhage—Time to Stop?

Catherine Mezzacappa, MD, MPH; Guadalupe Garcia-Tsao, MD

but who are otherwise compensated, are the most appropriate first population in whom to assess whether antibiotic prophylaxis can be safely withheld.

The findings of this systematic review and meta-analysis¹ highlight the paucity of high-quality evidence to support the practice of administering prophylactic broad-spectrum antibiotics to patients with cirrhosis presenting with upper gas-

trointestinal hemorrhage. The current level of evidence is also inadequate to answer whether it is time to stop this practice, which has become the standard of care. Modern trials of shorter duration and no antibiotic prophylaxis should be designed in specific patient populations to compare sequelae of antibiotic prophylaxis, including subsequent infections and all-cause mortality.

